Solutions to STATISTICS WORKSHEET-1

1. True.
2. Central Limit Theorem
3. Modeling bounded count data
4. All of the mentioned
5. Poisson
6. False
7. Hypothesis
8. 0
9. Outliers cannot conform to the regression relationship

10. A normal distribution is a bell-shaped frequency distribution curve. Most of the data values in a normal distribution tend to cluster around the mean. The further a data point is from the mean, the less likely it is to occur. There are many things, such as intelligence, height, and blood pressure, that naturally follow a normal distribution. The statistical term for it is **Gaussian distribution**.
11. Missing data can be dealt with in a variety of ways. I believe the most common reaction is to ignore it. Choosing to make no decision, on the other hand, indicates that your statistical programme will make the decision for you.
    The application will remove things in a listwise sequence most of the time. Depending on why and how much data is gone, listwise deletion may or may not be a good idea.
    Another common strategy among those who pay attention is imputation. Imputation is the process of substituting an estimate for missing values and analysing the entire data set as if the imputed values were the true observed values.
    There are some of the most prevalent methods to estimate it:

- Mean imputation

Calculate the mean of the observed values for that variable for all non-missing people.It has the advantage of maintaining the same mean and sample size, but it also has a slew of drawbacks. Almost all of the methods described below are superior to mean imputation.

- Substitution
  Assume the value from a new person who was not included in the sample.To put it another way, pick a new subject and employ their worth instead.
- Hot deck imputation
  A value picked at random from a sample member who has comparable values on other variables.To put it another way, select all the sample participants who are comparable on other factors, then choose one of their missing variable values at random.
  One benefit is that you are limited to just feasible values. In other words, if age is only allowed to be between 5 and 10 in your research, you will always obtain a value between 5 and 10.Another factor is the random element, which introduces some variation. For exact standard errors, this is crucial.
- Cold deck imputation
  A value picked deliberately from an individual with similar values on other variables.In most aspects, this is comparable to Hot Deck, but without the random variance. As an example, under the same experimental condition and block, you can always select the third individual.
- Regression imputation
  The result of regressing the missing variable on other factors to get a predicted value.As a result, instead of utilising the mean, you're relying on the anticipated value, which is influenced by other factors. This keeps the associations between the variables in the imputation model, but not the variability around the anticipated values.
- Stochastic regression imputation
  The predicted value of a regression plus a random residual value.This has all of the benefits of regression imputation plus the random component's benefits.The majority of multiple imputation is based on stochastic regression imputation.
- Interpolation and extrapolation

An estimate based on other observations made by the same person. It generally only works with data that is collected over time.Proceed with caution, though. For a variable like height in children–one that cannot be reduced through time–interpolation would make more sense. Extrapolation entails estimating beyond the data's true range, which necessitates making more assumptions than is necessary.

- Single or Multiple Imputation
  Single and multiple imputation are the two forms of imputation. When people say imputation, they usually mean single.
  The term "single" refers to the fact that you only use one of the seven methods to estimate the missing number outlined above.
  It's popular since it's simple to understand and generates a sample with the same number of observations as the complete data set.
  When listwise deletion eliminates a considerable amount of the data set, single imputation appears to be a tempting option. It does, however, have certain restrictions.
  Unless the data is Missing Completely at Random, certain imputation processes, such as means, correlations, and regression coefficients, result in skewed parameter estimations. The bias is frequently worse than with listwise deletion, which is most software's default.
  The level of the bias is determined by a number of factors, including the imputation technique, the missing data mechanism, the fraction of missing data, and the information in the data set.
  Furthermore, standard errors are underestimated by all single imputation approaches.Because the imputed observations are estimates, their values have a random error associated with them. However, your programme is unaware of this when you enter that estimate as a data point. As a result, it ignores the additional source of error, resulting in too-small standard errors and p-values.
  And, while imputation is straightforward in theory, it is difficult to master in reality. As a result, it isn't perfect, although it may suffice in some circumstances.
- As a result of multiple imputation, numerous estimates are generated. In multiple imputation, two of the approaches indicated above–hot deck and stochastic regression–work as the imputation method.

- The multiple estimates varied significantly because these two approaches contain a random component. This reintroduces some variance that your program can account for in order to provide reliable standard error estimates for your model.
- About 20 years ago, multiple imputation was a big advance in statistics. It eliminates many (but not all) difficulties with missing data and, when done correctly, leads to unbiased parameter estimations and accurate standard errors.

12. A/B testing in its simplest sense is an experiment on two variants to see which performs better based on a given metric. Typically, two consumer groups are exposed to two different versions of the same thing to see if there is a significant difference in metrics like sessions, click-through rate, and/or conversions. Using the visual above as an example, we could randomly split our customer base into two groups, a control group and a variant group. Then, we can expose our variant group with a red website banner and see if we get a significant increase in conversions. It's important to note that all other variables need to be held constant when performing an A/B test. Getting more technical, A/B testing is a form of statistical and two-sample hypothesis testing. **Statistical hypothesis testing** is a method in which a sample dataset is compared against the population data. **Two-sample hypothesis testing** is a method in determining whether the differences between the two samples are statistically significant or not.
13. Mean imputation is typically considered terrible practice since it ignores feature correlation. Consider the following scenario: we have a table with age and fitness scores, and an eight-year-old has a missing fitness score. If we average the fitness scores of people between the ages of 15 and 80, the eighty-year-old will appear to have a significantly greater fitness level than he actually does. Second, mean imputation decreases the variance of our data while increasing bias. As a result of the reduced variance, the model is less accurate and the confidence interval is narrower.
14. Linear regression is a basic and commonly used type of predictive analysis. The overall idea of regression is to examine two things: (1) does a set of predictor variables do a good job in predicting an outcome (dependent) variable? (2) Which variables in particular are significant predictors of the outcome variable, and in what way do they–indicated by

the magnitude and sign of the beta estimates–impact the outcome variable? These regression estimates are used to explain the relationship between one dependent variable and one or more independent variables. The simplest form of the regression equation with one dependent and one independent variable is defined by the formula y = c + b*x, where y = estimated dependent variable score, c = constant, b = regression coefficient, and x = score on the independent variable. Naming the Variables. There are many names for a regression's dependent variable. It may be called an outcome variable, criterion variable, endogenous variable, or regressand. The independent variables can be called exogenous variables, predictor variables, or regressors. Three major uses for regression analysis are (1) determining the strength of predictors, (2) forecasting an effect, and (3) trend forecasting.

15. There are two major branches of statistics.

Descriptive statistics:

This is a branch of statistics which deals with methods of collection of data, its presentation and organization in various forms, such as distribution tables, graphs (e.g., ogive, Lorenz curves, etc.), diagrams (e.g., pie charts) and finding measures of central tendency and measures of dispersion or spread which are used in the description of data. Managers, CEOs. etc. make use of descriptive statistics in presenting their annual reports, financial accounts and bank statements. Descriptive statistics is used to present the data in an understandable way, so that a meaningful description can be made.

Inferential or predictive statistics:

This is a branch of statistics which deals with techniques used for analysis of data, making estimates that lead to predictions and drawing conclusions or inferences from. limited information taken on sample basis and testing the reliability of the estimates or predictions.
Inferential statistics is used to make comparisons or predictions about a larger group, known as population, using information gathered about a small part of that population called a sample. Inferential statistics answers questions, such as "what is this data telling us about?" and "what should we

do?" Techniques used are forecasting trends, hypothesis testing, kurtosis, skewness, etc.